

# Distinct Substrings

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Time limit: 1.00 s   Memory limit: 512 MB

Count the number of distinct substrings that appear in a string.

## Input

The only input line has a string of length  $n$  that consists of characters a–z.

## Output

Print one integer: the number of substrings.

## Constraints

- $1 \leq n \leq 10^5$

## Example

Input:

abaa

Output:

8

Explanation: the substrings are a, b, aa, ab, ba, aba, baa and abaa.